

C programming questions

Basic operators and math functions only

1. Write a program to create a int, float, char, double type variables. Use scanf to take their values as input and print them in order. Before scanf for every variable, write a print statement to write "enter an int value : "....
2. Write a program which takes as input a float value (celcius temperature), it should output the temperature in kelvin and fahrenheit scale
3. Write a program to take input a principal amount (P), rate of interest (r in %) and time in years and it should print the final Amount along with the simple interest
4. Write a program which takes as input a double value which is an angle in degrees. It should use math library functions to print the sin, cos, tan, cosec, sec, cot values for that angle.
You should also apply inverse to all the sin, cos values you obtained to get the initial angle in degrees back and print them
5. Write a program whose input is principal amount, rate of interest in % and some number Target amount (T). now assume you are putting your principal amount in a bank which gives you compound interest yearly with rate of interest 'r'. How many years will you have to wait till your bank amount reaches your target amount. Calculate and output (hint : you'll have to use log function in the math library)
6. Consider the following program

```
#include<stdio.h>
int main(){
    char a,b;
    int x;
    scanf("%d",&x);
    scanf("%c",&a);
    scanf("%c",&b);
    printf("%d %c %c", x,a,b);
}
```

What is wrong with this program? See after running it will not run as you expected when you give integer, hit enter and give another char and hit enter

7. Write a program whose input is a principal amount (P), rate of interest 'r' in %, no. of years 'n'.
Your job is to output the total amount after 2n years. Where for first years you will add simple interest and from 2nd n years you will add compound interest
(simple interest will not change the principal amount, it wil add same ptr/100 every year.
Whereas for compound interest, the interest will add to your principal amount every year)
8. Write a program to swap 2 int values
(i) with using a temporary variable
(ii) without using a temporary variable
9. Write a program to print the last digit of a number and the last 2nd digit